

SIMATS ENGINEERING

ASSESSMENT TOOL 1 - High (INDUSTRY PROBLEM SOLVING TASKS)

COURSE CODE: CSA0921

COURSE NAME: Programming in Java

COURSE OUTCOME COVERED

CO4: *Design and develop GUI-based user interfaces using Abstract Window Toolkit, Swing, and Applets by implementing event handling, layout managers, AWT controls, and menus. (BL3,BL5)*

Assessment Tool Weightage: 10%

QUESTIONS: 1

Total Marks: 50

ANNEXURE A

INDUSTRY PROBLEM SOLVING TASKS

Code of Conduct:

/ B.SIVARANGJENII

(192521375)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

B.SIVARANGJENII

ANNEXURE

INDUSTRY PROBLEM SOLVING TASKS

1. Inventory Management System (BTL: BL3 – Apply)

Develop a GUI-based inventory system using Swing to manage stock levels, product details, and supplier information. Use appropriate controls like text fields, buttons, and tables. Implement event handling for adding, updating, and deleting records. Validate inputs such as product ID and quantity. Apply layout managers to ensure a clean and responsive interface. Analyze performance when handling large inventories and suggest improvements.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis

Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation
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CODE:

```

import java.applet.Applet;
import java.awt.*;
import java.awt.event.*;
import javax.swing.*;
import javax.swing.table.DefaultTableModel;

public class InventoryApplet extends Applet implements ActionListener {
    JTextField id, name, qty, supplier;
    JButton add, update, delete, clear;
    JTable table;
    DefaultTableModel model;

    public void init() {
        setLayout(new BorderLayout());
        // Fields
        id = new JTextField();
        name = new JTextField();
        qty = new JTextField();
        supplier = new JTextField();
        JPanel p = new JPanel(new GridLayout(4, 2));

        p.add(new JLabel("Product ID"));
        p.add(id);
        p.add(new JLabel("Product Name"));
        p.add(name);

```

```

p.add(new JLabel("Quantity"));
p.add(qty);
p.add(new JLabel("Supplier"));
p.add(supplier);
// Buttons
add = new JButton("Add");
update = new JButton("Update");
delete = new JButton("Delete");
clear = new JButton("Clear");
JPanel b = new JPanel();
b.add(add);
b.add(update);
b.add(delete);
b.add(clear);
// Table
model = new DefaultTableModel(
    new String[] {"ID", "Name", "Quantity", "Supplier"}, 0);
table = new JTable(model);
add(p, BorderLayout.NORTH);
add(new JScrollPane(table), BorderLayout.CENTER);
add(b, BorderLayout.SOUTH);
add.addActionListener(this);
update.addActionListener(this);
delete.addActionListener(this);
clear.addActionListener(this);
}
public void actionPerformed(ActionEvent e) {
    // ADD
    if (e.getSource() == add) {
        if (id.getText().isEmpty() ||
            name.getText().isEmpty() ||
            qty.getText().isEmpty() ||
            supplier.getText().isEmpty()) {
            JOptionPane.showMessageDialog(this,

```

```

        "All fields are required");
    return;
}
try {
    int q = Integer.parseInt(qty.getText());
    if (q < 0) {
        JOptionPane.showMessageDialog(this,
            "Quantity cannot be negative");
        return;
    }
    model.addRow(new Object[]{
        id.getText(),
        name.getText(),
        q,
        supplier.getText()
    });
    clearFields();
} catch (NumberFormatException ex) {
    JOptionPane.showMessageDialog(this,
        "Quantity must be a number");
}
}

// UPDATE
if (e.getSource() == update) {
    int r = table.getSelectedRow();
    if (r >= 0) {
        model.setValueAt(id.getText(), r, 0);
        model.setValueAt(name.getText(), r, 1);
        model.setValueAt(qty.getText(), r, 2);
        model.setValueAt(supplier.getText(), r, 3);
        clearFields();
    } else {
        JOptionPane.showMessageDialog(this,
            "Select a row");
    }
}

```

```

    }
}

// DELETE
if (e.getSource() == delete) {
    int r = table.getSelectedRow();
    if (r >= 0)
        model.removeRow(r);
    else
        JOptionPane.showMessageDialog(this,
            "Select a row");
}

// CLEAR
if (e.getSource() == clear)
    clearFields();
}

void clearFields() {
    id.setText("");
    name.setText("");
    qty.setText("");
    supplier.setText("");
}
}

```

OUTPUT:

Applet Viewer: InventoryApplet.class

Applet

Product ID

Product Name

Quantity

Supplier

ID	Name	Quantity	Supplier
P101	LAPTOP	10	ABC SUPPLIERS
PC109	MOBILES	25	ABC SUPPLIERS

Add

Update

Delete

Clear

Applet started

30°C

Partly cloudy

Search

ENG

IN

9%

21:09

20-08-2020